20/07/26

PCA continued + TSNE

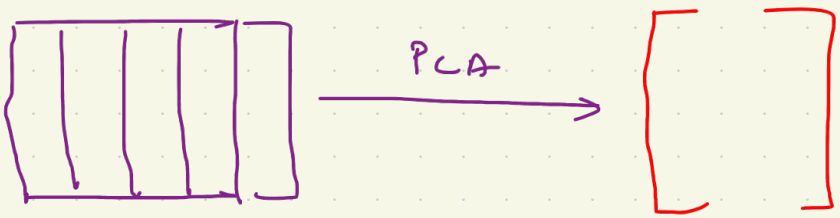
* Agenda ⇒

- PCA Recap / + code
- Maths
- limitations + use cases *
- t-SNE intuition ⇒
- t-SNE code ⇒

high level →

* Recap ⇒

* 5 dim^n data



Practical

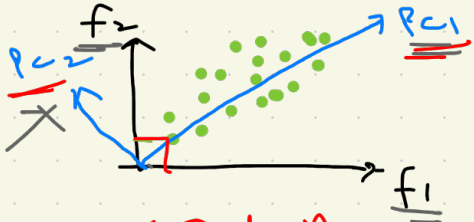
Total = 5 PCs

PC1	PC2	PC3	PC4	PC5
40 %	35 %	25 %	0	0

⇒ 3 PCs ⇒

PC3 & PC5

angle ?

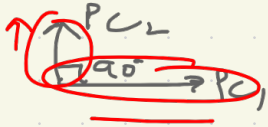


→ 90°

* 2 dimn

PC1

orthogonal 3 dimn



PCA → linear algo

→ linear combo

Ques

MC *

exist after PCA ?

f1 + f2

PC2 ⟹ linear combo

Yes ⟹ ✓✓

interpretation

wts ⟹ PCA

Age Year → Car price

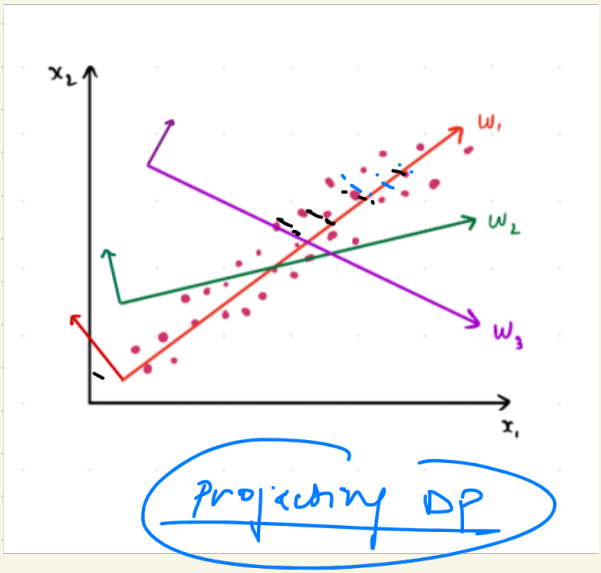
Age = -1 · Year + 2026

MC

Lin Reg^n → DOUBT

w_1 w_2 ⇒ eq^n

eq → DOUBT



⇒ Reg

Reg

MC → lin Reg

projecting DP

finding var.

monthly spend

+ve

Avg order value

PCA

IoT sensor

61 → 2

ML

4 10

VIF

10%

t-SNE

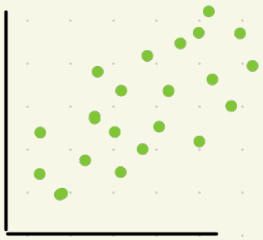
90%

Overall engagement

spending intensity

Maths ⇒ ∝

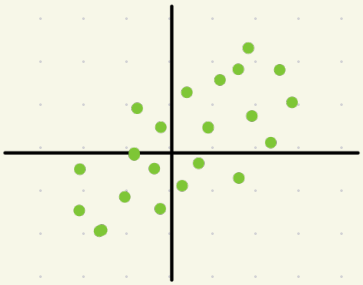
← feature scaling



Std.

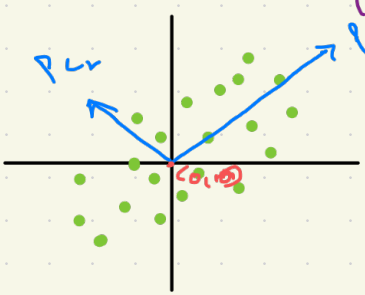
→ mean centring

→ var :- 1



Scaled DATA

$\vec{u}$ ⇒ Eigenvector



PC1 : $w^T x + w_0 = 0$

$w_0 = 0$

$w^T x = 0 \Rightarrow$

$\vec{u}^T x = 0$ $\vec{u}$ ×

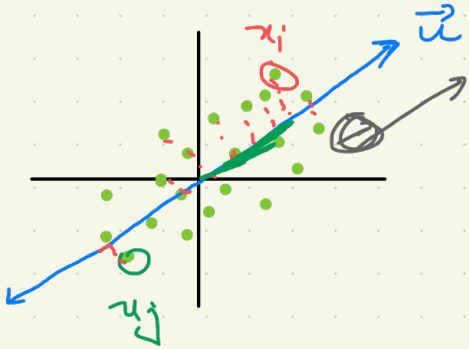
ML

2 dim°

$\vec{u} = \begin{bmatrix} 2.5 \\ 1.8 \end{bmatrix}$

eigen vector





$\text{Proj}_{\vec{u}} x_i = \frac{x_i \cdot u}{\|u\|} = 3.7$

$3.7 - 2.1$

$\text{Proj}_{\vec{u}} x_j = \frac{x_j \cdot u}{\|u\|} = -2.1$

$\neq 0$

$$\max \left[\frac{1}{n} \sum_{i=1}^{n} \frac{x_i \cdot u}{\|u\|} \right]$$

Soln

$\vec{u}$

$$\max_{\vec{u}} \left[\frac{1}{n} \sum_{i=1}^{n} \frac{(x_i \cdot u)^2}{\|u\|} \right]$$

constraint $\|u\| = 1$

$\|u\|^2 = 1$

constraint opt. problem

$(\sqrt{w_1^2 + w_2^2} = 1 \Rightarrow$

Lagrange multiplier

↓

Unconstraint

$\|u\| = 1$

$\|u\| - 1 = 0$

$$\max_{\vec{u}, \lambda} \frac{1}{n} \sum (x_i \cdot u)^2 + \lambda (\|u\|^2 - 1)$$

$\Rightarrow$ Lagrange multiplier

Matrix for ML

$A^T A = A^T = \|A\|^2$

$$\max_{\vec{u}} (X \cdot \vec{u})^2 + \lambda (\|u\|^2 - 1)$$

$\frac{1}{n}$

$(AB)^T = B^T \cdot A^T$

$$\frac{(X \cdot u)^T (X \cdot u)}{n} + \lambda (u^T u - 1)$$

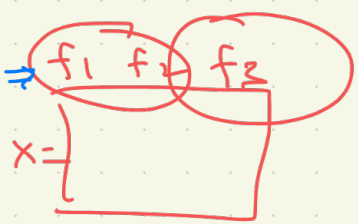
$$\max_{u,\lambda} \left[\frac{u^T \cdot X^T \cdot X \cdot u}{n} + \lambda(u^T u - 1) \right]$$

u → vector, λ → scalar ⇒ Covariance matrix

OBJ. func^n

Performance

efficiency



⇒ cov (X)

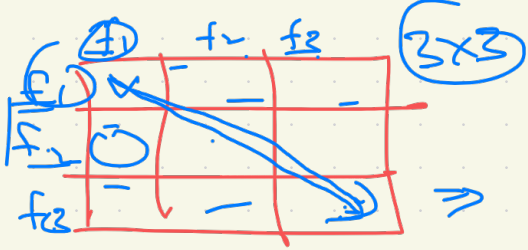
var(f_1, f_2)

var(f_1, f_3)

var(f_2, f_3) ⇒

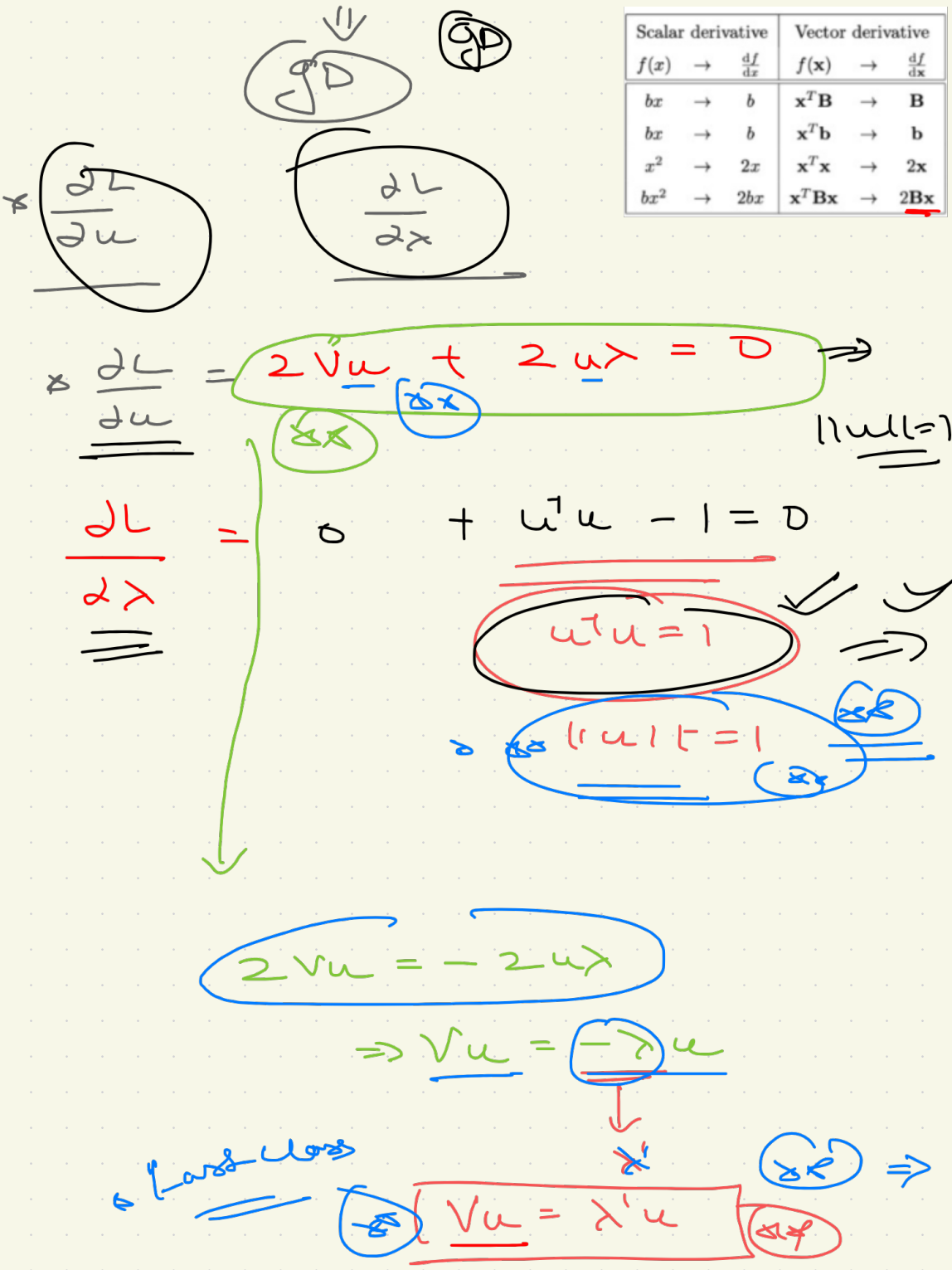


cov ⇒



$$\mathcal{L} = \max_{u,\lambda} \left[u^T V \cdot u + \lambda u^T u - \lambda \right]$$

u → vector, λ → scalar


Scalar derivative
$f(x)$ → $\frac{df}{dx}$
bx → b
bx → b
x^2 → $2x$
bx^2 → $2bx$
Vector derivative
$f(\mathbf{x})$ → $\frac{df}{d\mathbf{x}}$
$\mathbf{x}^T\mathbf{B}$ → $\mathbf{B}$
$\mathbf{x}^T\mathbf{b}$ → $\mathbf{b}$
$\mathbf{x}^T\mathbf{x}$ → $2\mathbf{x}$
$\mathbf{x}^T\mathbf{B}\mathbf{x}$ → $2\mathbf{B}\mathbf{x}$
$\frac{\partial L}{\partial u}$
$\frac{\partial L}{\partial x}$
$\frac{\partial L}{\partial u} = 2Vu + 2u\lambda = 0$
$\|u\|=1$
$\frac{\partial L}{\partial \lambda} = 0 + u^T u - 1 = 0$
$u^T u = 1$
$\|u\|_2 = 1$
$2Vu = -2u\lambda$
$\Rightarrow Vu = -\lambda u$
Lagrange
$Vu = \lambda' u$

Eigen value decomposition

V = matrix
u = vector
λ = scalar value

d × d

d

V → shape = d × d

$\vec{u}$ ⇒ d × 1

(d × d) (d × 1)

$\vec{u}$

$\vec{u}$, λ → np.linalg.eig(V)

[d]

[[ ]]

d frames → d

→ d

Conclude

i/p data

X
(n, d)

PCA →

X - new
(n, 2)
→ 2

1. Scaling ✓

2. Cov. matrix ⇒ V ⇒ $\frac{X^T X}{n}$

3. $\vec{u}$ & λ for V

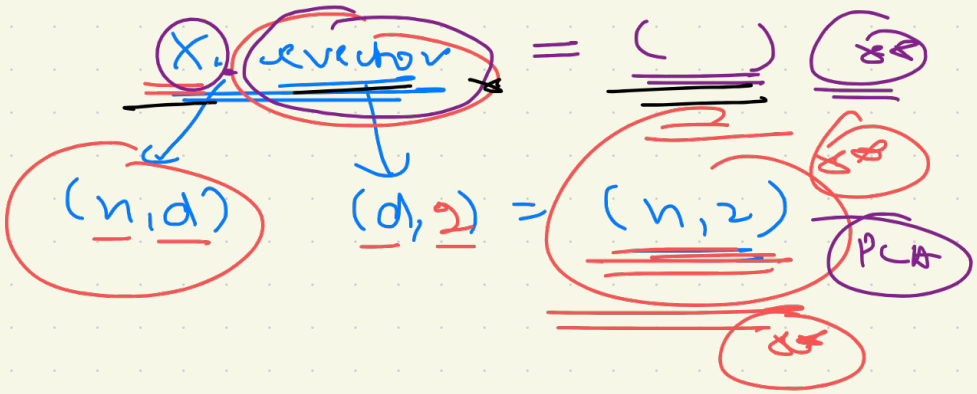
evec, eval = np.linalg.eig(V)

d → d

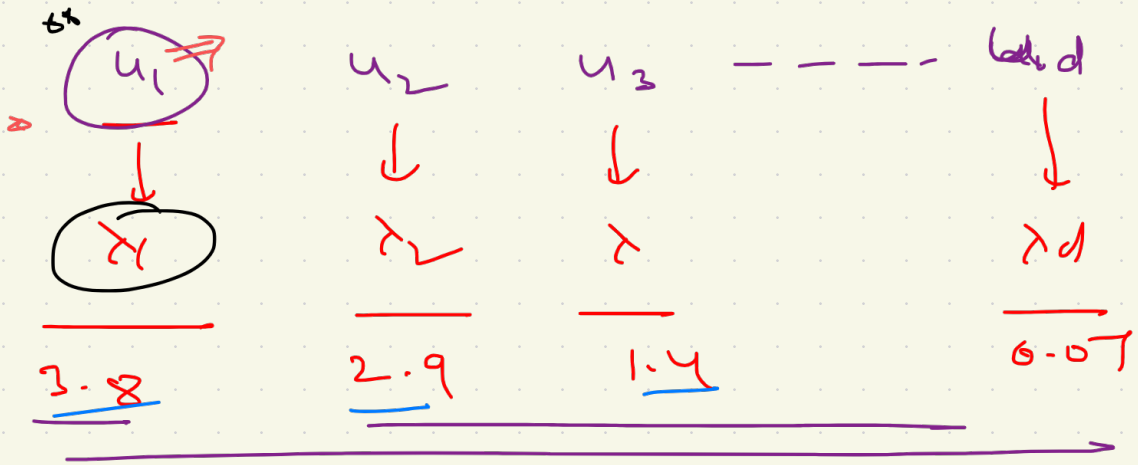
PC1 PC2 - - - - PCd

X

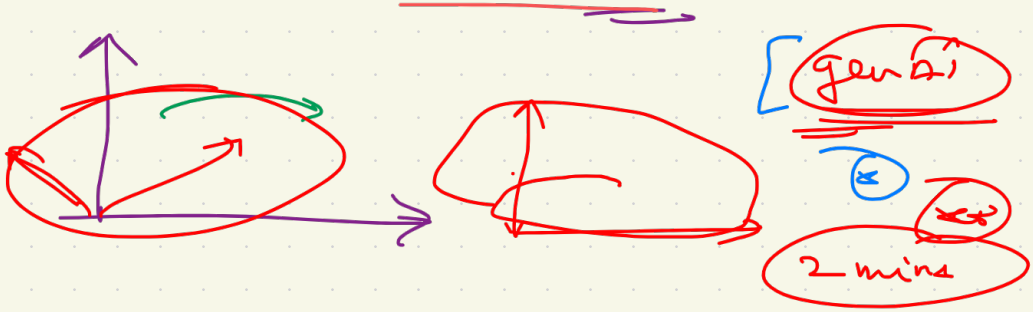
⇒ d, 2

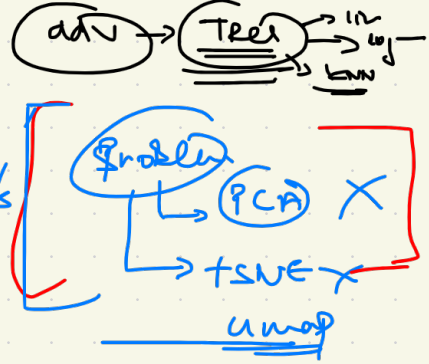


importance of PCs



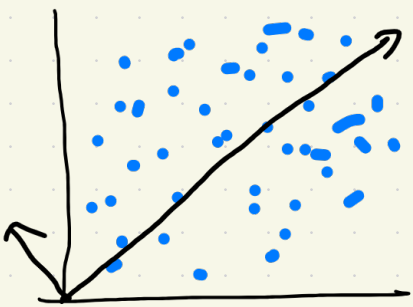
$$imp(u_1) = \frac{3.8}{\text{sum of all } \lambda s}$$





Assumptions

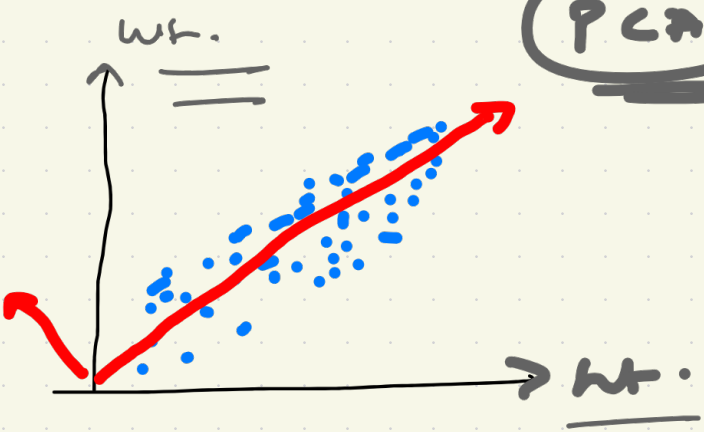
PCA ⇒ helping ?



Collinearity

X

PCA

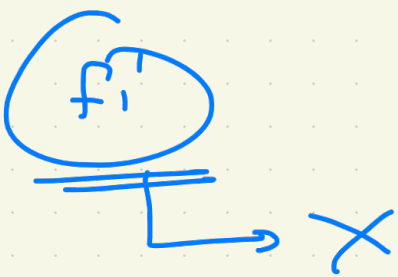


Limitations

1. linear algo ✗

2. low interpretability of PCs

$PC_1 = 0.8 \text{ mileage} + 1.4 \text{ odor} + \ldots$



3. Trade off — Blw info loss & dim^n Reduction

4. PCA → Not Robust to outliers
   - impacted By outliers

t-SNE

Maths ML

t-distributed stochastic Neighbourhood embedding

t-SNE

2008 → DL

used for viz^n

2D or 3D

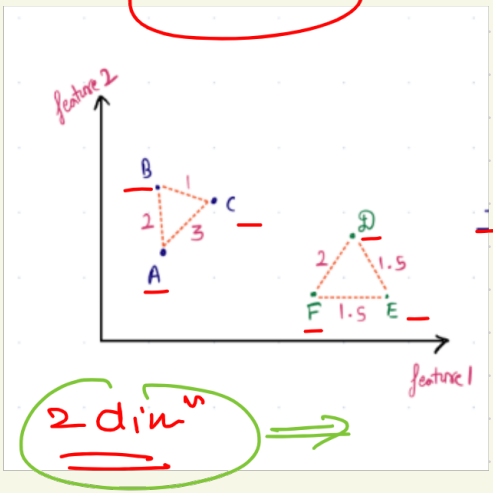
LLMs → embeddings

5000

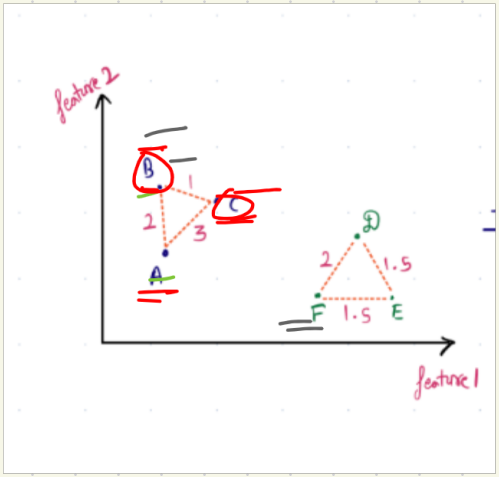
lower

t-SNE

KNN



2 dim^n



(i, j)

$Prob_N(i, j)$

Higher

↓

$Prob(i, j)$

lower dim

$Prob(i, j)$ ≈ $Prob(i, j)$

Higher — lower dim

Core principle

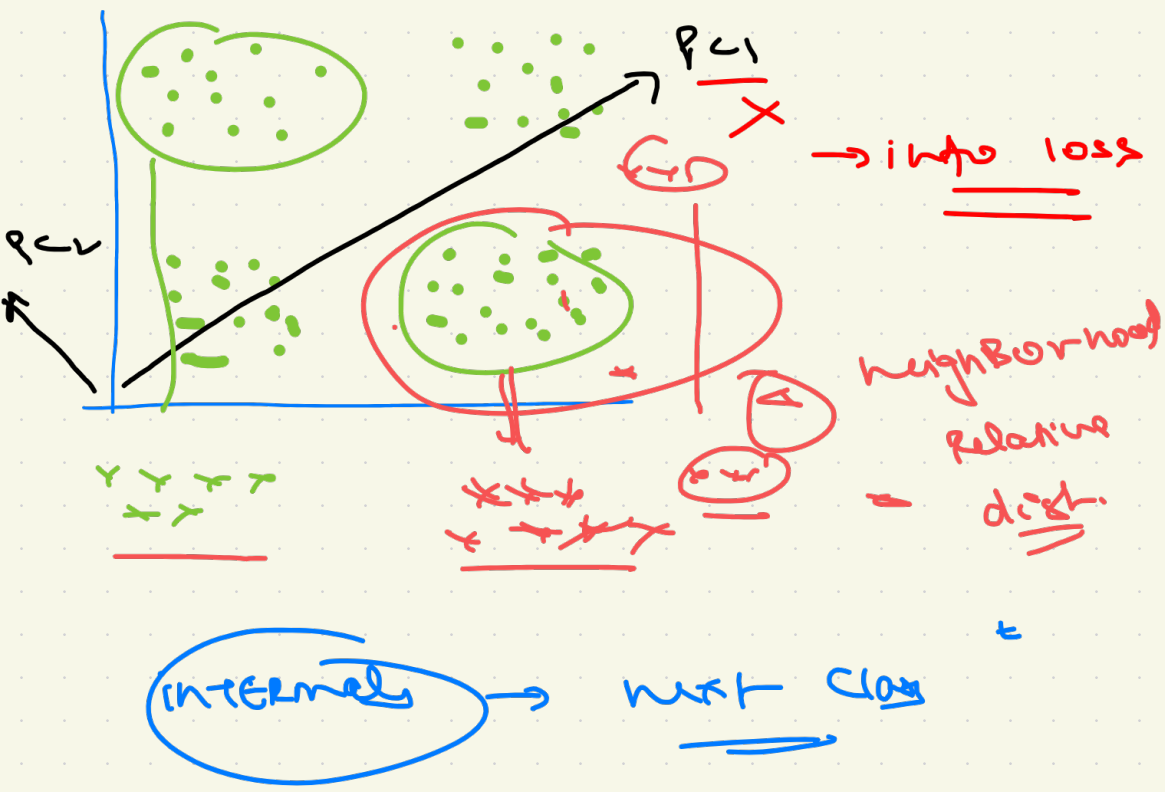
$Prob(i, j)$ ≈ $Prob(i, j)$

Higher — approx — lower dim

PCA ⟹ neighBorhood ✗

✗ Relative distances

→ max variance dir°



INTERNALLY → next class



DOUBT SESSION